Question #1 of 46

Question ID: 1674878

Which of the following risks is the *most* qualitative in nature?

A) Liquidity risk.

B) Reputational risk.

C) Credit risk.

Question #2 of 46

Question ID: 1674850

Which of the following forms of analysis has a backward-looking bias?

A) Quantitative tools.

B) Both quantitative tools and qualitative assessments.

C) Qualitative assessments.

Question #3 of 46

Question ID: 1674864

In addition to setting the overall risk tolerance, where are additional risk parameters such as limits, requirements, and guidelines codified?

A) The risk policy manual.

B) The IPS.

C) The risk framework.

Question #4 of 46

Question ID: 1674867

In relation to institutional investors with an explicit active risk budget in public equities, which of the following is *most accurate* of the desired investment strategy?

A) Allocate to growth and value factors.

B) Allocate to momentum factors.

C) Allocate to stock pickers.

Question #5 of 46

Question ID: 1674849

Which of the following are two important aspects of illiquid asset classes?

A) Zero cash flows and return unsmoothing issues.

B) Lack of cash flows and return smoothing issues.

C) Uncertain cash flows and return smoothing issues.

Valentina Flores is an alternative investment analyst with a sovereign wealth fund (SWF). The investment committee has recently asked for further information regarding illiquid real estate and private equity holdings and whether the SWF is reflecting an accurate picture of the related risk exposures. As a result, Felipe Pereyra, the chief investment officer, asks Flores to report on the accuracy of reported asset correlations and return volatility.

Pereyra then asks Flores to clarify four statements she has made in her report that relate to financial risks faced by institutional investors.

Statement 1:	"SWFs that are classified as savings funds share very similar objectives to endowments and foundations."
Statement 2:	"Funding liquidity refers to how quickly an asset can be sold at a fair price."
Statement 3:	"Mean variance optimization (MVO) is an appropriate optimization tool for institutions only after applying unsmoothing adjustments."
Statement 4:	"Conditional value at risk (CVaR) and Monte Carlo simulation (MCS) both deal with long-term risk measurement."

Question #6 - 9 of 46

Question ID: 1674854

Regarding the impact of the illiquid holdings on the reported correlations within the SWF, correlations will *most likely* be:

A) correctly stated.

B) overstated.

C) understated.

Question #7 - 9 of 46

Question ID: 1674855

Regarding the impact of the illiquid holdings on the reported return volatility within the SWF, return volatility will *most likely* be:

A) overstated.

B) correctly stated.

C) understated.

Question #8 - 9 of 46

Question ID: 1674856

Which of the following statements made by Flores in her report to the investment committee is accurate?

A) Statement 2 only.

B) Both Statements 1 and 2.

C) Statement 1 only.

Question #9 - 9 of 46

Question ID: 1674857

Which of the following statements made by Flores in her report to the investment committee is accurate?

A) Statement 3 only.

B) Statement 4 only.

C) Neither Statement 3 nor 4.

Question #10 of 46

Question ID: 1674852

Return smoothing results from investments in illiquid assets held for long periods of time in private markets. Which of the following statements regarding smoothing is *most accurate*?

A) The impact of smoothing is always an upward bias on return volatility.

B) The impact of smoothing may be either a downward or upward bias on return volatility.

C) The impact of smoothing is always a downward bias on return volatility.

Question #11 of 46

Question ID: 1674870

Which of the following is not a step in the enterprise risk management process?

A) Mitigate all risks.

B) Strategic analysis.

C) Reports and communications.

Chyou Zhao is an infrastructure analyst with Shulin Capital (SC), a long-term institutional investor. Ah Lam Huang is the head of infrastructure at SC, and is a member of the SC Investment Committee. As part of a new liquidity initiative, the committee is reviewing the SC contingency liquidity plan in the event of a market downturn.

Huang asks Zhao to compare direct and indirect forms of investment structures currently used by SC to invest in illiquid infrastructure assets over the medium to long term. Huang asks Zhao to focus particularly on the liquidity, diversification, and level of control of each investment structure.

Zhao identifies four illiquid infrastructure investments, each having a 12-to-15-year investment time horizon. Zhao studies each project and identifies the following characteristics:

- Project A has a higher level of liquidity.
- Project B has lower management costs.
- Project C has a higher level of diversification.
- Project D has an increased risk of additional liabilities.

Question #12 - 15 of 46

Question ID: 1674885

Which of the following infrastructure projects is *most likely* to provide a lower level of control over the investment?

A) Project B.

B) Project A.

C) Project C.

Question #13 - 15 of 46

Question ID: 1674886

Which of the following infrastructure projects is likely to be the *most* illiquid?

A) Project C.

B) Project D.

C) Project B.

Question #14 - 15 of 46

Question ID: 1674887

Which of the following projects is *most likely* to be an indirect infrastructure investment?

A) Project A, Project B, and Project C.

B) Project C only.

C) Project A and Project B.

Question #15 - 15 of 46

Question ID: 1674888

Which of the following projects is *most likely* to result in reduced principal-agent issues?

A) Project D only.

B) Project B and Project D.

C) Project A, Project B, and Project D.

Question #16 of 46

Question ID: 1674874

Which of the following considers the impacts of climate change mitigation and adaptation on local communities and individuals in developing countries, providing communications and support to those impacted?

A) Fair transition.

B) Just transition.

C) Equitable transition.

Question #17 of 46

Question ID: 1674882

Which of the following is an advantage of a direct investment rather than an indirect investment structure when investing in long-term illiquid assets?

A) Lower principal-agent issues.

B) Lower liability risks.

C) Lower staff costs.

Thomas Nixon has recently joined a new investment management firm as a senior risk manager. His first task is to make recommendations on the firm's risk management process in terms of using top-down and/or bottom-up approaches.

Nixon knows that the firm's institutional investors usually perform risk factor analysis to better understand their risk exposures. His next task is to do some research to determine what risks are likely to be overlooked by institutional investors and to prevent them from being overlooked.

In analyzing environmental and social risks faced by institutional investors, Nixon has been tasked to examine the firm's clients, who invest in real estate assets in coastal areas of Europe. Some of those coastal areas are affected by rising sea levels, with real estate asset values falling and properties becoming uninsurable.

Nixon's final task for the day is to look at climate risk responses—specifically, strategies to reduce the vulnerability of investors to both physical and transition climate risks.

Question #18 - 21 of 46

Question ID: 1674859

The firm's risk management process will *most likely* be successful using:

A) a top-down perspective only.

B) both a top-down and bottom-up perspective.

C) a bottom-up perspective only.

Question #19 - 21 of 46

Question ID: 1674860

Which of the following risks is *most likely* to be overlooked by institutional investors?

A) Asset allocation drift.

B) Inflation.

C) Liquidity.

Question #20 - 21 of 46

Question ID: 1674861

The climate-related risk pertaining to the coastal areas of Europe is *best* described as:

A) chronic physical climate risk.

B) climate transition risk.

C) acute physical climate risk.

Question #21 - 21 of 46

Question ID: 1674862

In the context of climate risk responses, a carbon tax is *best* described as:

A) a reduction strategy.

B) a mitigation strategy.

C) an adaptation strategy.

TOPIC: CASES IN RISK MANAGEMENT: INSTITUTIONAL

TOTAL POINT VALUE OF THIS QUESTION SET IS 12 POINTS

Greendale Capital (GC) is a midsized hedge fund with assets of $600 million, attracting funds from green investors in the expanding ESG marketplace. Mike Stone, a senior analyst with GC, is evaluating three potential illiquid investments, each with a 7-to-10-year investment horizon.

Coastal Conference Centers (CCC) is an up-and-coming luxury brand that has made a name for itself delivering superior corporate events that incorporate coastal activities. CCC plans to expand its prime coastal property portfolio and build upon its successful business model. A major driver of this success was the early decision to ensure its business operations are carbon neutral. The adoption of the latest solar and wind power technology has enabled CCC to bring in new green clients and carve out a loyal and growing customer base. Stone is considering a direct investment of $8 million in CCC.

Neon Carbon Capture (NCC) is at the forefront of carbon capture technologies, which help fossil fuel–reliant companies adapt to meet greener carbon regulations. Carbon capture as a methodology is still evolving and has yet to be widely accepted as a mainstream solution due to its high implementation costs. Stone is considering an indirect investment of $4 million in NCC.

CruGas Refineries (CGR) is an established name in the oil-refining sector. CGR has both downstream and upstream investment projects and is looking to raise a direct capital investment of $6 million to invest in the new technology required to adapt its business model. Stone is considering investing $3 million in CGR.

Stone's sister, Prue Silverton, is an ESG analyst with the FREIDA pension plan, a large institutional investor. When discussing the FREIDA ESG approach with her brother, Silverton explains that FREIDA is a universal owner.

Question #22 - 25 of 46

Question ID: 1674890

When evaluating the liquidity risk to GC of accepting the following two investments, which of the following is *most accurate*?

A) The liquidity risk to GC is relatively high in CCC and low in NCC.

B) The liquidity risk to GC is relatively high in both CCC and NCC.

C) The liquidity risk to GC is relatively low in both CCC and NCC.

Question #23 - 25 of 46

Question ID: 1674891

Which of the following is *most likely* to be the biggest risk factor for CCC over the investment horizon?

A) Transition risk.

B) Chronic physical risk.

C) Acute physical risk.

Question #24 - 25 of 46

Question ID: 1674892

In evaluating the three investments, which is the *most* illiquid for GC and why?

A) NCC, due to the investment structure.

B) CGR, due to the additional sector risk.

C) CCC, due to the investment size.

Question #25 - 25 of 46

Question ID: 1674893

Based only on information provided in the scenario, which of the following is *most likely* to be accurate regarding the ESG exposures of the FREIDA pension plan?

A) Exposures tilted away from firms that are currently heavily dependent on fossil fuels.

B) Exposures tilted toward firms that will prosper in a carbon-neutral economy.

C) A similar level of exposure as the wider economy.

Question #26 of 46

Question ID: 1674868

Which of the following is the first step in the five-step liquidity risk management process?

A) Establish liquidity risk parameters.

B) Develop a cash flow model and project future expected cash flows.

C) Review the enterprise risk framework.

Question #27 of 46

Question ID: 1674869

Which of the following is the final step in the five-step liquidity risk management process?

A) Stress test liquidity needs and cash flow projections.

B) Develop a cash flow model and project future expected cash flows.

C) Develop an emergency plan.

Question #28 of 46

Question ID: 1674863

Which of the following is the *most* important investment decision made by the board?

A) Internal team or external managers.

B) Strategic asset allocation.

C) Required liquidity profile.

Question #29 of 46

Question ID: 1674848

Which of the following is inaccurate of risk systems?

A) Returns-based risk systems are cheaper than holdings-based risk systems.

B) Returns-based risk systems are not feasible for hedge fund investments.

C) Holdings-based risk systems are more difficult to implement than returns-based risk systems.

Question #30 of 46

Question ID: 1674875

In designing investment policy and strategy, which of the following is *most accurate* of long-term institutional investors?

A) Allocate to strike a balance between both fixed-income investments and riskier higher-return investments.

B) Allocate primarily to fixed income to reduce risk to the payments they were set up to provide.

C) Allocate primarily to risky and illiquid assets to provide high expected returns in the long run.

Question #31 of 46

Question ID: 1674851

Which of the following asset classes are available to meet liquidity needs in a crisis?

A) Private equity, real estate, infrastructure, and fixed income.

B) Real estate and fixed income.

C) Fixed income.

Question #32 of 46

Question ID: 1674873

Which of the following is not categorized as a climate mitigation opportunity?

A) Water treatment.

B) Batteries and storage.

C) Materials.

Chatsworth University (CU) is one of the oldest in England. The Chatsworth Endowment (CE), originally founded in 1867 by donations from the Chatsworth family estate, has grown to current assets under management of £4.2 billion. For many years, Edward Colquhoun, the long-standing CE chief investment officer, established a Norway model investment philosophy. However, since his retirement three years ago, Hilary Thatcher, the new CIO—together with the backing of the CE investment committee—has expanded holdings of alternatives and tilted toward the Canada model approach. In making these changes, CE has

expanded its internal investment team in the areas of private assets and infrastructure, recruiting Oliver Frankland and Alicia Yardley.

Before the upcoming investment committee meeting, Frankland and Yardley write a report proposing two new alternative direct investments in private equity. Each of the two projects requires a proposed investment of £25 million with a 20-year intended time horizon:

- Project A is a venture capital project investing in alternative heating technologies aiming to profit from the transition from gas heating and water systems to non–fossil fuel alternatives.
- Project B is a venture capital project investing in the development of commercial vehicle electric battery technologies.

Within the report, Frankland and Yardley estimate the speed of transition away from fossil fuels using scenario analysis by estimating a slower transition scenario, a midrange transition scenario, and a faster transition scenario based upon current industry transition estimates. To support the investment proposal, Frankland states that venture capital projects are very capital efficient due to capital being called for over time. Yardley states that capital calls are likely to slow in a crisis, providing countercyclical benefits for CE and some downside protection in a crisis.

Catherine Emmanuel, the newly appointed chief risk officer, has been evaluating the existing CE risk framework, largely still the legacy of Edward Colquhoun. As part of an initial audit of the risk framework, Emmanuel notes that different risk management techniques are used for different CE assets classes. Emmanuel reviews the CE risk tolerance statement established many years ago and notes that it is an asset-only approach rather than in asset-liability terms. In addition, she notes that CE views risk largely in silos with separate risk teams.

Emmanuel reviews the investment proposal of Frankland and Yardley and lists four risk issues for further investigation at the upcoming investment committee meeting:

- Issue 1—liquidity risk
- Issue 2—the denominator effect
- Issue 3—changed legal liabilities
- Issue 4—impact of a crisis

A few days later, Thatcher opens the investment committee meeting, during which the investment proposal from Frankland and Yardley is discussed.

Question #33 - 36 of 46

Question ID: 1674895

Thatcher asks Frankland and Yardley if they think liquidity risk will be highlighted as a significant risk for CE as a result of these two investments. Which of the following is *most accurate*?

A) Yes; both are direct investments in private equity.

B) Yes; both are long-term illiquid investments.

C) No; they represent relatively small allocations.

Question #34 - 36 of 46

Question ID: 1674896

In relation to the statement made by Yardley regarding capital calls, which of the following is *most accurate*?

A) Yardley is correct; capital calls are likely to slow down.

B) Yardley is incorrect; capital calls are unlikely to be impacted.

C) Yardley is incorrect; capital calls are likely to speed up.

Question #35 - 36 of 46

Question ID: 1674897

At the investment committee meeting, Catherine Emmanuel explains her risk concerns in further detail. Which of Emmanuel's concerns is *most likely* to be incorrect?

A) The denominator effect increases the risk of both projects in a market crisis.

B) As both are direct investments, CE has increased liability risk from the projects.

C) As both are direct investments, CE has increased liquidity risk from the projects.

Question #36 - 36 of 46

Question ID: 1674898

Which of the following risk concerns raised by Catherine Emmanuel is *most likely* to require action by CE?

A) CE has adopted an asset-only risk tolerance statement.

B) CE adopts a silo approach to risk management.

C) CE uses different risk methodologies for different asset classes.

Question #37 of 46

Question ID: 1674872

Which organization mandates that banks, asset managers, and asset owners use scenario analysis to estimate the risks and opportunities of climate change and make appropriate disclosures?

A) Neither the Intergovernmental Panel on Climate Change nor the Task Force on Climate-Related Disclosures.

B) The Task Force on Climate-Related Disclosures.

C) The Intergovernmental Panel on Climate Change.

Question #38 of 46

Question ID: 1674879

Which of the following is a commonly overlooked risk by institutional investors?

A) Currency risk.

B) Inflation risk.

C) Credit risk.

Question #39 of 46

Question ID: 1674871

Which of the following is not a risk measure estimating losses beyond a value-at-risk (VaR) estimate?

A) Conditional VaR.

B) Unexpected tail loss.

C) Expected shortfall.

Question #40 of 46

Question ID: 1674866

Which of the following *most accurately* describes the ultimate risk consideration of institutional investors?

A) To maximize the risk-return tradeoff.

B) To be able to meet contractual payments.

C) To optimize the risk-return tradeoff.

Question #41 of 46

Question ID: 1674876

You are conducting due diligence before investing in a fund containing long-term illiquid assets. Which of the following is not critical to evaluate before investing as part of the due diligence process?

A) Track record of the brokerage firm.

B) Independence of the external auditors.

C) Quality of the non-executive directors.

Question #42 of 46

Question ID: 1674881

Which of the following is an advantage of an indirect form of investment in long-term illiquid assets over a direct investment approach?

A) There is greater liquidity.

B) It is easier to manage capital.

C) There is lower concentration risk.

Question #43 of 46

Question ID: 1674877

What is the main motivation behind pension funds and sovereign wealth funds building internal teams of merchant banking professionals?

A) Increase managerial control of projects.

B) Reduce investment management fees.

C) Increase liquidity via the timing of exits.

Question #44 of 46

Question ID: 1674865

With an external manager approach, which of the following is a good starting point in planning and developing liquidity parameters at the portfolio level?

A) The legal terms in place.

B) The liquidity horizon report.

C) The stress test results.

Question #45 of 46

Question ID: 1674880

Illiquid investments require institutional investors to consider how which of the following two risks interact?

A) Liquidity risk and market risk.

B) Credit risk and market risk.

C) Credit risk and liquidity risk.

Question #46 of 46

Question ID: 1674883

Which of the following is the main motivation for large pension funds and sovereign wealth funds to set up their own large team of merchant banking professionals experienced in making direct investments in illiquid asset classes?

A) To reduce concentration risks.

B) To increase liquidity.

C) To save on fund management fees.